Lecture 11: Pseudoinverses and Linear Equations

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🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Explicit solutions for consistent data
- Least-squares solutions for inconsistent data
- Minimum-norm solutions when $\|\mathbf{f}_{null}\| = 0$

📌 Before We Begin

Consider all possible cases based on ***data consistency*** and ***object nullity***:

Cases of the Data Space $\mathbb{V}$

- $\mathbb{V}$ consists only of $\mathbb{V}_{con}$ (trivial $\mathbb{V}_{incon}$): $R = M$ (*onto*, all consistent data).
  - Any $\mathbf{g} \in \mathbb{V}$ is always consistent: $\mathbf{g} = \mathbf{g}_{con}$.
- $\mathbb{V}$ consists of both $\mathbb{V}_{con}$ and a nontrivial $\mathbb{V}_{incon}$: $R < M$ (not *onto*).
  - Any $\mathbf{g} \in \mathbb{V}$ can be expressed as: $\mathbf{g} = \mathbf{g}_{con} + \mathbf{g}_{incon}$ (inconsistent data).
  - Some $\mathbf{g} \in \mathbb{V}$ may satisfy: $\mathbf{g} = \mathbf{g}_{con} + 0$ (consistent data).

Cases of the Object Space $\mathbb{U}$

- $\mathbb{U}$ consists only of $\mathbb{U}_{meas}$ (trivial $\mathbb{U}_{null}$): $R = N$ (*one-to-one*, no null space).
  - Any $\mathbf{f} \in \mathbb{U}$ is always measured: $\mathbf{f} = \mathbf{f}_{meas}$.
- $\mathbb{U}$ consists of both $\mathbb{U}_{meas}$ and a nontrivial $\mathbb{U}_{null}$: $R < N$ (not one-to-one).

- Any $\mathbf{f} \in \mathbb{U}$ can be expressed as: $\mathbf{f} = \mathbf{f}_{meas} + \mathbf{f}_{null}$.
- Some $\mathbf{f} \in \mathbb{U}$ may satisfy: $\mathbf{f} = \mathbf{f}_{meas} + 0$ (no null component).

Exact Soluition When $\mathbf{H}^{-1}$ Exists

For a *bijective* operator $\mathbf{H}$ ($R = M = N$), the inverse $\mathbf{H}^{-1}$ exists. The the estimate of $\hat{\mathbf{f}}$ is obtained as:

$$\hat{\mathbf{f}} = \mathbf{H}^{-1}\mathbf{g}.$$

This case corresponds to both trivial $\mathbb{V}_{incon}$ and trivial $\mathbb{U}_{null}$.

❓ What if $\mathbf{H}^{-1}$ does not exist?

Solving Linear Equations Using Pseudoinverses

The Moore-Penrose pseudoinverse provides a method **to solve systems of linear equations**, particularly when an exact solution may not exist due to **noise**, i.e., **inconsistent data**.

A real-world system can be expressed as:

$$g_m = \sum_{n=1}^{N} H_{mn} f_n + \epsilon_m, \quad m = 1, \ldots, M, \quad m\text{-th component}$$

$$\mathbf{g} = \mathbf{H}\mathbf{f} + \epsilon, \quad \text{Matrix-vector form}$$

where - $\epsilon \in \mathbb{R}^M$: Noise column vector with element ϵ_m, which is not characteristic of the object $\mathbf{f}$.
The goal is to find solutions to $\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$, i.e., to find an estimate of $\mathbf{f}$ (N unknown values $\{\hat{f}_n\}$ from M equations).

📝 Example: A System of Linear Equations

A system of three equations for two unknowns can be written as a matrix equation:

$$\begin{cases} x + y = 10 \\ 2x + y = 13 \\ x + 3y = 24 \end{cases} \iff \begin{bmatrix} 10 \\ 13 \\ 24 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

This shows how a system of linear equations can be represented using a matrix equation $\mathbf{g} = \mathbf{H}\mathbf{f}$, where $\mathbf{H}$ is an 3×2 matrix, $\mathbf{f} \in \mathbb{R}^2$ is an unknown vector, and $\mathbf{g} \in \mathbb{R}^3$ is the given data vector.

❓ Are all elements of the noise vector ϵ in $\mathbb{V}_{incon}$?

Existence of inconsistent data $\mathbf{g}_{incon}$ means $R < M$ and $\mathbb{V}_{incon}$ is non-trivial. The data $\mathbf{g}$ with random noise ϵ added can be expressed as:

$$\mathbf{g} = \sum_{m=1}^{M} \beta_m \mathbf{v}_m = \sum_{m=1}^{R} \beta_m \mathbf{v}_m + \sum_{m=R+1}^{M} \beta_m \mathbf{v}_m = \mathbf{g}_{con} + \mathbf{g}_{incon}$$

$$\epsilon = \sum_{m=1}^{M} \gamma_m \mathbf{v}_m = \sum_{m=1}^{R} \gamma_m \mathbf{v}_m + \sum_{m=R+1}^{M} \gamma_m \mathbf{v}_m = \epsilon_{con} + \epsilon_{incon}$$

$$\mathbf{g}_{con} = \sum_{m=1}^{R} \beta_m \mathbf{v}_m = \mathbf{H}\mathbf{f} + \epsilon_{con} = \mathbf{H}\hat{\mathbf{f}}, \quad \mathbf{g}_{incon} = \epsilon_{incon} = \sum_{m=R+1}^{M} \gamma_m \mathbf{v}_m$$

⭐ *Measurement noise ϵ lies in $\mathbb{V}$, because it is added to the data. It can be in both $\mathbb{V}_{con}$ and $\mathbb{V}_{incon}$.*

Existence and Uniqueness of Exact Solutions

Existence Theorems

The equation $\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$ has **at least one exact** solution for **every** $\mathbf{g}$ if and only if:

- Every $\mathbf{g}$ lies in the **column space** (**range**), i.e., it is a linear combination of the columns of $\mathbf{H}$; The columns of $\mathbf{H}$ span $\mathbb{V} = \mathbb{E}^M$;
- $R = M \leq N$;
- $\mathbf{H}\mathbf{H}^+$ is nonsingular;
- A right inverse of $\mathbf{H}$ exists;
- $\mathbb{V}_{incon}$ is trivial, every $\mathbf{g} \in \mathbb{V}_{con}$;
- $\mathbf{H}$ is *onto*.

The equation $\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$ has **at least one exact** solution for **a particular** $\mathbf{g}$ if and only if:

- The $\mathbf{g}$ lies in the **column space** (**range**), i.e., it is a linear combination of the columns of $\mathbf{H}$;
- $R < M \leq N$;
- $\mathbf{H}\mathbf{H}^+\mathbf{g} = \mathbf{g}$;
- $\mathbb{V}_{incon}$ is nontrivial, but the particular $\mathbf{g} \in \mathbb{V}_{con}$;

Uniqueness Theorems

Equation $\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$ has **at most one exact** solution for **every** $\mathbf{g}$ if and only if:

- The columns of $\mathbf{H}$ are linearly independent;
- $R = N \leq M$;
- $\mathbf{H}^{+}\mathbf{H}$ is nonsingular;
- A left inverse of $\mathbf{H}$ exists;
- $\mathbb{U}_{null}$ is trivial;
- $\mathbf{H}$ is *one-to-one*;

A **unique exact** solutions **exist** only if $\mathbf{H}$ is a **square** and **full rank** matrix ($R = M = N$).

Explicit Solutions for Consistent Data

Unique, Exact Soloutions

If $\mathbf{H}$ is a **square, full rank** matrix ($R = M = N$), the system $\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$ has a **unique** solution:

$$\mathbf{H}^{-1}\mathbf{g} = \mathbf{H}^{-1}\mathbf{H}\hat{\mathbf{f}}$$
$$\mathbf{H}^{-1}\mathbf{g} = \hat{\mathbf{f}}$$

Right Inverse

If $R = M$ (**full row rank**, but not necessarily square), $\mathbb{V}_{incon}$ is trivial, and a right inverse of $\mathbf{H}$ exists:

$$\mathbf{H}\mathbf{H}_{R}^{-1} = \mathbf{I}_{M}$$
$$\mathbf{H}\mathbf{H}^{\dagger}(\mathbf{H}\mathbf{H}^{\dagger})^{-1} = \mathbf{I}_{M}$$
$$\mathbf{H}_{R}^{-1} = \mathbf{H}^{\dagger}(\mathbf{H}\mathbf{H}^{\dagger})^{-1}$$

where,

- $\mathbf{H}_{R}^{-1}$: $N \times M$ right inverse of $\mathbf{H}$
- $\mathbf{I}_{M}$: $M \times M$ identity matrix

$$\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$$
$$\mathbf{I}_{M}\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$$
$$\mathbf{H}\mathbf{H}_{R}^{-1}\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$$
$$\mathbf{H}_{R}^{-1}\mathbf{g} = \hat{\mathbf{f}}$$

If $R = M$, then $\mathbf{H}_{R}^{-1} = \mathbf{H}^{+}$.

Data Consistency and Pseudoinverses

If $R < M$, $\mathbf{H}$ has *linearly dependent rows*, and $\mathbb{V}_{incon}$ is nontrivial.

However, even when $\mathbb{V}_{incon}$ is nontrivial, a particular data $\mathbf{g}$ may still be consistent, i.e., $\mathbf{H}\mathbf{H}^{+}\mathbf{g} = \mathbf{P}_{con}\mathbf{g} = \mathbf{g} \in \mathbb{V}_{con}$. In such case, the pseudoinverese provides an exact solution:

$$\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$$
$$\mathbf{P}_{con}\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$$
$$\mathbf{H}\mathbf{H}^{+}\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$$
$$\mathbf{H}^{+}\mathbf{g} = \hat{\mathbf{f}}$$

General Exact Solution

If $R < N$, $\mathbf{H}$ has a nontrivial $\mathbb{U}_{null}$. For consistent data, the general exact solution is:

$$\hat{\mathbf{f}} = \hat{\mathbf{f}}_{meas} + \hat{\mathbf{f}}_{null}$$
$$\hat{\mathbf{f}} = \mathbf{H}^{+}\mathbf{g} + \mathbf{P}_{null}\mathbf{y}$$
$$\hat{\mathbf{f}} = \mathbf{H}^{+}\mathbf{g} + [\mathbf{I}_N - \mathbf{H}^{+}\mathbf{H}]\mathbf{y}$$

where

- $\mathbf{y} \in \mathbb{U}$: Arbitrary vector

🧠 *Recall the projector operator onto the null space* (in Lecture 10): $\mathcal{P}_{null} = \mathcal{I}_{\mathbb{U}} - \mathcal{H}^{+}\mathcal{H}$

Multiplying both sides by $\mathbf{H}$, we can check whether this is the solution to $\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$:

$$\mathbf{H}\hat{\mathbf{f}} = \mathbf{H}\mathbf{H}^{+}\mathbf{g} + [\mathbf{H}\mathbf{I}_N - \mathbf{H}\mathbf{H}^{+}\mathbf{H}]\mathbf{y}$$
$$\mathbf{H}\hat{\mathbf{f}} = \mathbf{I}_M\mathbf{g} + \mathbf{0}\mathbf{y} = \mathbf{g}.$$

Therefore, $\hat{\mathbf{f}} = \mathbf{H}^{+}\mathbf{g} + [\mathbf{I}_N - \mathbf{H}^{+}\mathbf{H}]\mathbf{y}$ is the solution to $\mathbf{g} = \mathbf{H}\hat{\mathbf{f}}$.

🧠 *Recall the Penrose equation 1* (in Lecture 10): $\mathcal{H}\mathcal{H}^{\sharp}\mathcal{H} = \mathcal{H}$

❓ What if the data is inconsistent?

Least-Squares Solutions for Inconsistent Data

If $R < M$ and the $\mathbf{g}$ is inconsistent, **no exact solution exists**.

A **least-squares solution** minimizes the squared $\mathbb{L}_2$ norm of the **residual** vector $\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}$, which quantifies how much a particular estimate $\hat{\mathbf{f}}$ deviates from reproducing the data under the action of $\mathbf{H}$:

$$\hat{\mathbf{f}}_{LS} = \arg\min_{\hat{\mathbf{f}}} \|\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}\|^2$$

Derivation of the Normal Equation

Minimizing the squared $\mathbb{L}_2$ norm of the residual vector implies that its derivatives with respect to the optimal $\hat{\mathbf{f}}$ vanishes: $\frac{\partial}{\partial \hat{\mathbf{f}}}\|\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}\|^2 = 0$.

$$\begin{aligned}
\frac{\partial}{\partial \hat{\mathbf{f}}}\|\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}\|^2 &= \frac{\partial}{\partial \hat{\mathbf{f}}}\left[\mathbf{g}^\dagger\mathbf{g} - \mathbf{g}^\dagger\mathbf{H}\hat{\mathbf{f}} - (\mathbf{H}\hat{\mathbf{f}})^\dagger\mathbf{g} + (\mathbf{H}\hat{\mathbf{f}})^\dagger\mathbf{H}\hat{\mathbf{f}}\right] \\
&= \frac{\partial}{\partial \hat{\mathbf{f}}}\left[\mathbf{g}^\dagger\mathbf{g} - \mathbf{g}^\dagger\mathbf{H}\hat{\mathbf{f}} - \hat{\mathbf{f}}^\dagger\mathbf{H}^\dagger\mathbf{g} + \hat{\mathbf{f}}^\dagger\mathbf{H}^\dagger\mathbf{H}\hat{\mathbf{f}}\right] \\
&= -\mathbf{g}^\dagger\mathbf{H} + \hat{\mathbf{f}}^\dagger\mathbf{H}^\dagger\mathbf{H} = \left[-\mathbf{H}\mathbf{g}^\dagger + \mathbf{H}^\dagger\mathbf{H}\hat{\mathbf{f}}^\dagger\right]^T = 0 \\
\mathbf{H}^\dagger\mathbf{H}\hat{\mathbf{f}} &= \mathbf{H}^\dagger\mathbf{g} \quad \text{Normal equation}
\end{aligned}$$

A similar result is obtained by differentiating with respect to $\hat{\mathbf{f}}^\dagger$:

$$\begin{aligned}
\frac{\partial}{\partial \hat{\mathbf{f}}^\dagger}\|\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}\|^2 &= \frac{\partial}{\partial \hat{\mathbf{f}}^\dagger}\left[\mathbf{g}^\dagger\mathbf{g} - \mathbf{g}^\dagger\mathbf{H}\hat{\mathbf{f}} - (\mathbf{H}\hat{\mathbf{f}})^\dagger\mathbf{g} + (\mathbf{H}\hat{\mathbf{f}})^\dagger\mathbf{H}\hat{\mathbf{f}}\right] \\
= \frac{\partial}{\partial \hat{\mathbf{f}}^\dagger}\left[\mathbf{g}^\dagger\mathbf{g} - \mathbf{g}^\dagger\mathbf{H}\hat{\mathbf{f}} - \hat{\mathbf{f}}^\dagger\mathbf{H}^\dagger\mathbf{g} + \hat{\mathbf{f}}^\dagger\mathbf{H}^\dagger\mathbf{H}\hat{\mathbf{f}}\right] &= -\mathbf{H}^\dagger\mathbf{g} + \mathbf{H}^\dagger\mathbf{H}\hat{\mathbf{f}} = 0 \\
\mathbf{H}^\dagger\mathbf{H}\hat{\mathbf{f}} &= \mathbf{H}^\dagger\mathbf{g} \quad \text{Normal equation}
\end{aligned}$$

Solving the normal equation $\mathbf{H}^\dagger\mathbf{H}\hat{\mathbf{f}} = \mathbf{H}^\dagger\mathbf{g}$ is equivalent to **finding a least-squares solution** to the original equation.

📝 Calculating the Residual Norm (For Reference)

The following derivation shows how the residual norm can be explicitly calculated.

$$\begin{aligned}
\|\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}\|^2 = \langle \mathbf{g} - \mathbf{H}\hat{\mathbf{f}}, \mathbf{g} - \mathbf{H}\hat{\mathbf{f}}\rangle &= (\mathbf{g} - \mathbf{H}\hat{\mathbf{f}})^\dagger(\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}) = (\mathbf{g}^\dagger - (\mathbf{H}\hat{\mathbf{f}})^\dagger)(\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}) \\
&= \mathbf{g}^\dagger\mathbf{g} - \mathbf{g}^\dagger\mathbf{H}\hat{\mathbf{f}} - (\mathbf{H}\hat{\mathbf{f}})^\dagger\mathbf{g} + (\mathbf{H}\hat{\mathbf{f}})^\dagger\mathbf{H}\hat{\mathbf{f}} \\
&= \|\mathbf{g}\|^2 - \langle \mathbf{g}, \mathbf{H}\hat{\mathbf{f}}\rangle - \langle \mathbf{H}\hat{\mathbf{f}}, \mathbf{g}\rangle + \|\mathbf{H}\hat{\mathbf{f}}\|^2 \\
&= \|\mathbf{g}\|^2 - \langle \mathbf{H}\hat{\mathbf{f}}, \mathbf{g}\rangle^* - \langle \mathbf{H}\hat{\mathbf{f}}, \mathbf{g}\rangle + \|\mathbf{H}\hat{\mathbf{f}}\|^2 \\
&= \|\mathbf{g}\|^2 - \left[\mathrm{Re}\left(\langle \mathbf{H}\hat{\mathbf{f}}, \mathbf{g}\rangle\right) - \mathrm{Im}\left(\langle \mathbf{H}\hat{\mathbf{f}}, \mathbf{g}\rangle\right)\right] - \left[\mathrm{Re}\left(\langle \mathbf{H}\hat{\mathbf{f}}, \mathbf{g}\rangle\right) + \mathrm{Im}\left(\langle \mathbf{H}\hat{\mathbf{f}}, \mathbf{g}\rangle\right)\right] + \|\mathbf{H}\hat{\mathbf{f}}\|^2 \\
&= \|\mathbf{g}\|^2 - 2\mathrm{Re}(\langle \mathbf{H}\hat{\mathbf{f}}, \mathbf{g}\rangle) + \|\mathbf{H}\hat{\mathbf{f}}\|^2
\end{aligned}$$

🧠 *Recall the inner product property* (in Lecture 2): $\langle \mathbf{u}, \mathbf{v}\rangle = \langle \mathbf{v}, \mathbf{u}\rangle^*$

Existence and Uniqueness of Least-Squares Solutions

- **Existence**: $\hat{\mathbf{f}}$ is estimated where the $\mathbb{L}_2$ norm of the residual vector is 0 or some minimum value, which means there is **always at least one solution**.

$$\hat{\mathbf{f}}_{LS} = \arg\min_{\hat{\mathbf{f}}} \|\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}\|^2, \quad \mathbf{H}^\dagger\mathbf{H}\hat{\mathbf{f}} = \mathbf{H}^\dagger\mathbf{g}$$

- **Uniqueness**
  - If $R = N$, $\mathbb{U}_{null}$ is trivial, and a left inverse of $\mathbf{H}^\dagger\mathbf{H}$ exists, ensuring a **unique solution**.

$$\mathbf{H}^\dagger\mathbf{H}\hat{\mathbf{f}}_{LS} = \mathbf{H}^\dagger\mathbf{g}$$
$$(\mathbf{H}^\dagger\mathbf{H})^{-1}\mathbf{H}^\dagger\mathbf{H}\hat{\mathbf{f}}_{LS} = (\mathbf{H}^\dagger\mathbf{H})^{-1}\mathbf{H}^\dagger\mathbf{g}$$
$$\hat{\mathbf{f}}_{LS} = (\mathbf{H}^\dagger\mathbf{H})^{-1}\mathbf{H}^\dagger\mathbf{g} = \mathbf{H}^+\mathbf{g}$$

  - If $R < N$, $\mathbb{U}_{null}$ is nontrivial, and the least-squares solution is not unique.

📝 Calculating $(\mathbf{H}^\dagger\mathbf{H})^{-1}\mathbf{H}^\dagger$ (For Reference)

$$\mathbf{H}^\dagger\mathbf{H} = \sum_{k=1}^{R} \sqrt{\mu_k}\mathbf{u}_k\mathbf{v}_k^\dagger \sum_{n=1}^{R} \sqrt{\mu_n}\mathbf{v}_n\mathbf{u}_n^\dagger$$
$$= \sum_{k=1}^{R}\sum_{n=1}^{R} \sqrt{\mu_k}\sqrt{\mu_n}\mathbf{u}_k\mathbf{v}_k^\dagger\mathbf{v}_n\mathbf{u}_n^\dagger = \sum_{k=1}^{R}\sum_{n=1}^{R} \sqrt{\mu_k}\sqrt{\mu_n}\mathbf{u}_k\delta_{kn}\mathbf{u}_n^\dagger$$
$$= \sum_{k=1}^{R} \mu_k\mathbf{u}_k\mathbf{u}_k^\dagger$$

If $R = N$, then $\mathbf{H}^\dagger\mathbf{H}$ is invertible, and its inverse is:

$$(\mathbf{H}^\dagger\mathbf{H})^{-1} = \sum_{k=1}^{N} \frac{1}{\mu_k}\mathbf{u}_k\mathbf{u}_k^\dagger$$

Multiplying by $\mathbf{H}^\dagger$:

$$(\mathbf{H}^\dagger\mathbf{H})^{-1}\mathbf{H}^\dagger = \sum_{k=1}^{N} \frac{1}{\mu_k}\mathbf{u}_k\mathbf{u}_k^\dagger \sum_{m=1}^{N} \sqrt{\mu_m}\mathbf{u}_m\mathbf{v}_m^\dagger = \sum_{k=1}^{N} \frac{\sqrt{\mu_k}}{\mu_k}\mathbf{u}_k\mathbf{v}_k^\dagger$$
$$= \sum_{k=1}^{N} \frac{1}{\sqrt{\mu_k}}\mathbf{u}_k\mathbf{v}_k^\dagger = \mathbf{H}^+$$

General Least-Squares Solutions

If $R < N$, $\mathbf{H}$ has a nontrivial $\mathbb{U}_{null}$.

$$\hat{\mathbf{f}}_{LS} = \mathbf{H}^{+}\mathbf{g} + [\mathbf{I}_N - \mathbf{H}^{+}\mathbf{H}]\mathbf{y}$$

where:

- $[\mathbf{I}_N - \mathbf{H}^{+}\mathbf{H}]\mathbf{y}$: The most general vector in $\mathbb{U}_{null}$.

Multiplying both sides by $\mathbf{H}^{\dagger}\mathbf{H}$, we can check whether this satisfies the normal equation:

$$\mathbf{H}^{\dagger}\mathbf{H}\hat{\mathbf{f}}_{LS} = \mathbf{H}^{\dagger}\mathbf{H}\mathbf{H}^{+}\mathbf{g} + [\mathbf{H}^{\dagger}\mathbf{H}\mathbf{I}_N - \mathbf{H}^{\dagger}\mathbf{H}\mathbf{H}^{+}\mathbf{H}]\mathbf{y}$$
$$= \mathbf{H}^{\dagger}\mathbf{g} + [\mathbf{H}^{\dagger}\mathbf{H} - \mathbf{H}^{\dagger}\mathbf{H}]\mathbf{y} = \mathbf{H}^{\dagger}\mathbf{g}$$

Thus, the general least-squares solution satisfies the normal equation $\mathbf{H}^{\dagger}\mathbf{H}\hat{\mathbf{f}} = \mathbf{H}^{\dagger}\mathbf{g}$.

🧠 *Recall the pseudoinverse property* (in Lecture 10): $[\mathcal{H}^{\dagger}\mathcal{H}]\mathcal{H}^{+} = \mathcal{H}^{\dagger}$

Minimum-Norm Solutions

A **minimum-norm** solution minimizes the norm of $\hat{\mathbf{f}}$. Since $\|\hat{\mathbf{f}}_{null}\| \geq 0$, the solution is minimized when $\|\hat{\mathbf{f}}_{null}\| = 0$.

$$\|\hat{\mathbf{f}}\|^2 = \|\hat{\mathbf{f}}_{meas} + \hat{\mathbf{f}}_{null}\|^2 \leq \|\hat{\mathbf{f}}_{meas}\|^2 + \|\hat{\mathbf{f}}_{null}\|^2$$
$$= \|\hat{\mathbf{f}}_{meas}\|^2 + \|\hat{\mathbf{f}}_{null}\|^2 = \|\mathbf{H}^{+}\mathbf{g}\|^2 + \|[\mathbf{I}_N - \mathbf{H}^{+}\mathbf{H}]\mathbf{y}\|^2$$
$$\|\hat{\mathbf{f}}\|^2 = \|\mathbf{H}^{+}\mathbf{g}\|^2$$

🧠 *Recall the norm properties* (in Lecture 2): $\|\mathbf{u}\| \geq 0$ and $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

Minimum-norm ($\|\hat{\mathbf{f}}_{null}\| = 0$) **exact** solution for consistent data

$$\hat{\mathbf{f}}_{MN} = \mathbf{H}^{+}\mathbf{g}$$

Minimum-norm ($\|\hat{\mathbf{f}}_{null}\| = 0$) **least-squares** solution for inconsistent data

$$\hat{\mathbf{f}}_{MNLS} = \mathbf{H}^{+}\mathbf{g}$$

Either case, the **minimum-norm** ($\|\hat{\mathbf{f}}_{null}\| = 0$) solution is $\mathbf{H}^{+}\mathbf{g}$.

Explicit Minimum-Norm Solutions in the SVD Domain

Consider an arbitrary data vector $\mathbf{g}$ and an arbitrary estimate of the object $\hat{\mathbf{f}}$.

$$\mathbf{g} = \sum_{k=1}^{M} \beta_k \mathbf{v}_k, \quad \beta_k = \mathbf{v}_k^\dagger \mathbf{g}, \quad \hat{\mathbf{f}} = \sum_{k=1}^{N} \hat{\alpha}_k \mathbf{u}_k, \quad \hat{\alpha}_k = \mathbf{u}_k^\dagger \mathbf{f}$$

The explicit form of the minimum-norm estimates is:

$$\begin{aligned}\hat{\mathbf{f}} &= \hat{\mathbf{f}}_{MN} = \hat{\mathbf{f}}_{MNLS} = \mathbf{H}^+ \mathbf{g} \\ &= \sum_{k=1}^{R} \frac{1}{\sqrt{\mu_k}} \mathbf{u}_k \mathbf{v}_k^\dagger \sum_{m=1}^{R} \beta_m \mathbf{v}_m = \sum_{k=1}^{R} \sum_{m=1}^{M} \frac{\beta_m}{\sqrt{\mu_k}} \mathbf{u}_k \mathbf{v}_k^\dagger \mathbf{v}_m \\ &= \sum_{k=1}^{R} \sum_{m=1}^{M} \frac{\beta_m}{\sqrt{\mu_k}} \mathbf{u}_k \delta_{km} = \sum_{k=1}^{R} \frac{\beta_k}{\sqrt{\mu_k}} \mathbf{u}_k\end{aligned}$$

🧠 *Recall the Moore-Penrose pseudoinverse* (in Lecture 10): $\mathcal{H}^+ = \sum_{k=1}^{R} \frac{1}{\sqrt{\mu_k}} \mathbf{u}_k \mathbf{v}_k^\dagger$

An algorithm to compute these estimates:

1. Perform an SVD on $\mathbf{H}$ and obtain its singular system $\{\mathbf{u}_k, \mathbf{v}_k, \mu_k\}$
2. Compute $\{\beta_k\}$ using the vectors $\{\mathbf{v}_k\}$: $\beta_k = \mathbf{v}_k^\dagger \mathbf{g}$
3. Compute $\{\alpha_k, k = 1, \ldots, R\}$: $\hat{\alpha}_k = \frac{\beta_k}{\sqrt{\mu_k}}$ (inverse filtering)
4. Compute $\hat{\mathbf{f}} = \sum_{k=1}^{N} \hat{\alpha}_k \mathbf{u}_k$, where $\hat{\alpha}_k = 0$ if $k > R$

💡 *SVD computation on large matrices is impractical. Instead, iterative algorithms can be used.*

Noise Amplification

A major drawback of pseudoinverse solutions is the noise (in $\mathbb{V}_{con}$) amplification for small μ_k.

$$\mathbf{g} = \mathbf{H}\mathbf{g} + \epsilon, \quad \epsilon = \sum_{k=1}^{M} \gamma_k \mathbf{v}_k, \quad \gamma_k = \epsilon^\dagger \mathbf{v}_k$$

$$\hat{\mathbf{f}}_{MN} = \hat{\mathbf{f}}_{MNLS} = \sum_{k=1}^{R} \left[\frac{\beta_k}{\sqrt{\mu_k}} + \frac{\gamma_k}{\sqrt{\mu_k}} \right] \mathbf{u}_k = \hat{\mathbf{f}}_{meas} + \sum_{k=1}^{R} \frac{\gamma_k}{\sqrt{\mu_k}} \mathbf{u}_k$$

If μ_k is close to zero, error can become significant.

⭐ *The pseudoinverse solutions discard* $\epsilon_{incon} = \sum_{m=R+1}^{M} \gamma_m \mathbf{v}_m$, *but* $\epsilon_{con} = \sum_{m=1}^{R} \gamma_m \mathbf{v}_m$ *can be amplified by* μ_k *close to zero.*

📌 Summary of Solutions

The type of solution depends on whether a specific $\mathbf{g}$ is **consistent** and whether the solution satisfies $\|\mathbf{f}_{null}\| = 0$.

- **Exact solutions** for consistent data
- **Least-squares solutions** for inconsistent data
- **Minimum norm solutions** when $\|\mathbf{f}_{null}\| = 0$
- General soltuions when $\|\mathbf{f}_{null}\| > 0$

Accordingly, the possible solutions are summarized as follows:

- Trivial $\mathbb{U}_{null}$, trivial $\mathbb{V}_{incon}$ $(R = N = M)$
  Unique and exact solution

$$\hat{\mathbf{f}} = \mathbf{H}^{-1}\mathbf{g}$$

- $\|\mathbf{f}_{null}\| = 0$, consistent data
  Minimum norm exact solution

$$\hat{\mathbf{f}} = \hat{\mathbf{f}}_{MN} = \mathbf{H}^{+}\mathbf{g}$$

- $\|\mathbf{f}_{null}\| = 0$, inconsistent data
  Minimum norm least-squares solution

$$\hat{\mathbf{f}}_{LS} = \hat{\mathbf{f}}_{MNLS} = \mathbf{H}^{+}\mathbf{g}$$

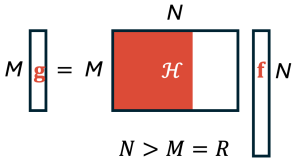
- $\|\mathbf{f}_{null}\| > 0$, consistent data
  General exact solution

$$\hat{\mathbf{f}} = \mathbf{H}^{+}\mathbf{g} + [\mathbf{I}_N - \mathbf{H}^{+}\mathbf{H}]\mathbf{y}$$

- $\|\mathbf{f}_{null}\| > 0$, inconsistent data
  General least-squares solution

$$\hat{\mathbf{f}}_{LS} = \mathbf{H}^{+}\mathbf{g} + [\mathbf{I}_N - \mathbf{H}^{+}\mathbf{H}]\mathbf{y}$$

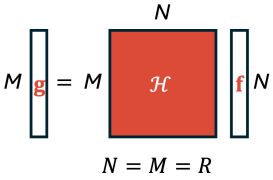
Underdetermined



$N > M = R$

$\text{nullity}(\mathcal{H}) = N - R$
Infinitely many exact solutions and
$\dim(\mathbb{V}_{incon}) = 0$
Always consistent, always exact solutions

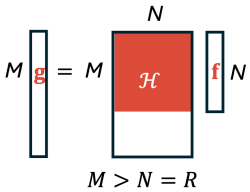
Square



$N = M = R$

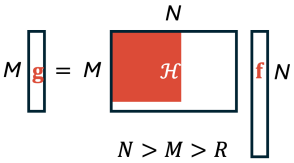
$\text{nullity}(\mathcal{H}) = 0$
Unique exact solution and
$\dim(\mathbb{V}_{incon}) = 0$
Always consistent, always exact solutions

Overdetermined

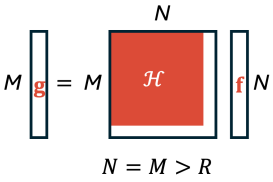


$M > N = R$

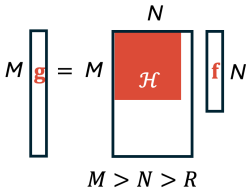
$\text{nullity}(\mathcal{H}) = 0$
Unique exact solutions or
$\dim(\mathbb{V}_{incon}) = M - R$
Can be inconsistent, no exact solutions



$N > M > R$



$N = M > R$



$M > N > R$

$\text{nullity}(\mathcal{H}) = N - R$ → Infinitely many exact solutions or
$\dim(\mathbb{V}_{incon}) = M - R$ → Can be inconsistent, no exact solutions